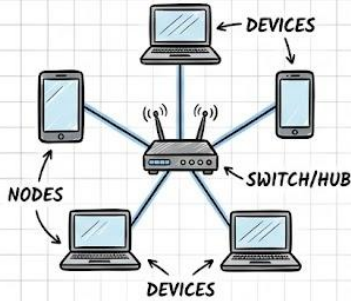


TASK-1

NETWORK TOPOLOGIES DIAGRAM & USE CASES

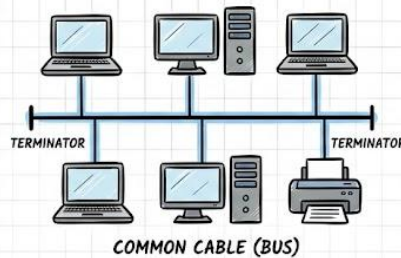
STAR TOPOLOGY



USE CASE: WhatsApp Messaging & Calls

A central server (switch) handles communication. If one client fails, others stay connected. Fast data routing.

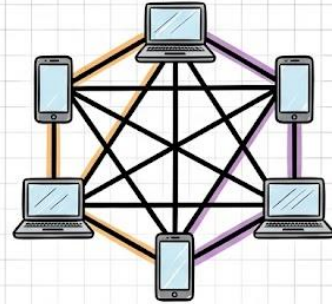
BUS TOPOLOGY



USE CASE: IPL Live Streaming

Broadcasting a single stream to multiple devices simultaneously. Efficient bandwidth usage for identical content delivery.

MESH TOPOLOGY (FULL)



USE CASE: Flipkart Order Processing (Multi-Warehouse)

Critical data paths between regional centers. Highly resilient; traffic reroutes immediately if one connection or node fails.

TASK-2

TOPOLOGY	PROS (4)	CONS (4)	BEST USE CASE	EXAMPLE APP
STAR	EASY SETUP, EASY TROUBLESHOOTING, FAST, RELIABLE	CENTRAL DEVICE FAILURE, MORE CABLES, COSTLY, DEPENDS ON SWITCH	HOME & OFFICE LAN	INSTAGRAM
BUS	LOW COST, LESS CABLE, SIMPLE SETUP, GOOD FOR SMALL NETWORKS	CABLE FAILURE STOPS NETWORK, SLOW WITH MANY DEVICES, HARD TO TROUBLESHOOT, LIMITED SIZE	SMALL LAN	PAYTM
RING	NO COLLISIONS, EQUAL ACCESS, STABLE	ONE FAILURE AFFECTS NETWORK, HARD	INDUSTRIAL NETWORKS	ZOMATO

TOPOLOGY	PROS (4)	CONS (4)	BEST USE CASE	EXAMPLE APP
MESH	PERFORMANCE, HANDLES TRAFFIC WELL	TO ADD DEVICES, DIFFICULT MAINTENANCE, EXPENSIVE		
	HIGHLY RELIABLE, MULTIPLE PATHS, SECURE, FAULT TOLERANT	VERY EXPENSIVE, COMPLEX SETUP, MORE CABLES, DIFFICULT MAINTENANCE	DATA CENTERS & BANKS	INSTAGRAM / PAYTM

TASK-3

SUITABLE TOPOLOGY: MESH TOPOLOGY

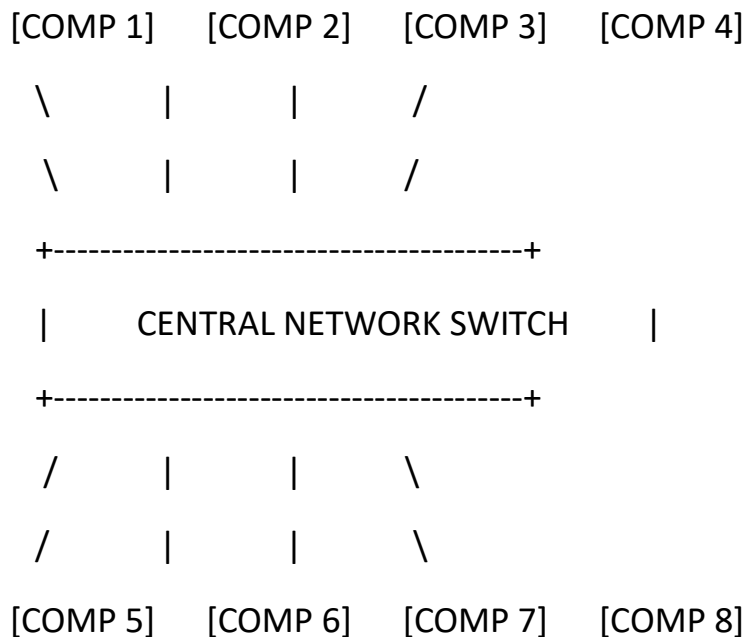
I WOULD CHOOSE **MESH TOPOLOGY** FOR THE GAME SERVERS BECAUSE IT PROVIDES MULTIPLE DATA PATHS, ENSURING HIGH RELIABILITY IF ONE CONNECTION FAILS. IT OFFERS FAST AND STABLE COMMUNICATION, REDUCING LAG DURING MULTIPLAYER MATCHES. ALTHOUGH IT IS MORE EXPENSIVE TO SET UP AND MAINTAIN, ITS SPEED AND RELIABILITY MAKE IT THE BEST CHOICE FOR ONLINE GAMES LIKE PUBG OR FREE FIRE.

TASK-4

RECOMMENDED TOPOLOGY: STAR TOPOLOGY

FOR AN 8-COMPUTER WAREHOUSE SETUP, **STAR TOPOLOGY** IS THE IDEAL CHOICE. ALL 8 COMPUTERS WILL CONNECT DIRECTLY TO A CENTRAL NETWORK SWITCH VIA ETHERNET CABLES.

NETWORK SKETCH



REASONING

- **HIGH RELIABILITY & EASY TROUBLESHOOTING:** IF ONE WAREHOUSE COMPUTER OR CABLE FAILS DURING BUSY ORDER PROCESSING, THE OTHER 7 COMPUTERS REMAIN FULLY OPERATIONAL, KEEPING FULFILLMENT ACTIVE.
- **LOW COST & SIMPLE SETUP:** A SINGLE 8-PORT OR 16-PORT UNMANAGED SWITCH AND STANDARD CAT6 CABLES MAKE THIS SETUP EXTREMELY AFFORDABLE AND STRAIGHTFORWARD TO INSTALL.
- **SEAMLESS SCALABILITY:** AS INVENTORY OPERATIONS GROW, YOU CAN EASILY PLUG NEW COMPUTERS INTO AVAILABLE SWITCH PORTS WITHOUT SHUTTING DOWN OR INTERRUPTING THE EXISTING NETWORK.